

Real-Time Possessing Relationship Detection for Sports Analytics

Yinda Xu, Yonggang Peng

Zhejiang University, Hangzhou 310027, P. R. China
E-mail: {yinda_xu, pengyg}@zju.edu.cn

Abstract: In this paper, we propose a novel algorithm for relationship detection. This task involves the tracking of a target object and human pose. The target object is tracked with a visual object tracker. The human poses are estimated via a keypoint detector while the person identities are preserved with a simple yet effective IoU tracker. Finally, a possessing relationship inference is made based on the position information of the tracked target and humans. This algorithm meets the real-time requirement by running at over 20 FPS and we give an application illustration in sports analytics.

Key Words: Visual Object Tracking, Human Pose Estimation, Computer Vision, Deep Learning, Sports Analytics

1 Introduction

Machine learning techniques [2, 3] have been utilized in sports analytics by providing a statistical perspective for player performance or making predictions based on historical records. However, compared to analysis over structured data such as historical win rate or possession percentage, the analysis from video data remains challenging due to the unstructured nature of video data.

During the last decade, Deep Learning methods have been continuously pushing the performance of visual representation [4–6] to a higher stage. With deep visual representation, powerful vision systems have been emerging in numerous downstream vision tasks such as Object Detection [7], Semantic Segmentation [8], Face Recognition [9], Human Pose Estimation [10], to name but a few. Their progress allows us to have a finer-grained understanding of visual data captured in the real world and gives birth to many previously unseen applications.

Among these tasks, Human-Object Interaction (HOI) emerges recently and attracts much attention from the research community [11–13]. Gkioxari et al. proposed InteractNet [11] that gives a simple yet generic solution via extra prediction heads and multi-task learning. Fang et al. introduced part-level information [13] and largely improved the performance. Kato et al. proposed a solution by modeling the instance relationship with Graph Convolutional Networks (GCN) [12]. While researchers have been seeking for generic solutions that are compatible with current main stream academic benchmarks, demands from specific applications (e.g. analyzing sports games with possessing relationship information) do not receive much attention. However, it is usually specific applications that are involved in the sports analytics.

In this paper, we propose an algorithm framework for possessing relationship detection during sports games. This task involves a target object and an unspecified number of persons and aiming at detecting the possessing relationship in a video of sports games. In the first stage, the target position and state (size) are predicted with a visual object tracker [1] while keypoints of present persons are detected with a human pose estimator [10]. In the second stage, possessing relationship reasoning is performed based on the position in-

formation obtained during the first stage. Person identities (IDs) are tracked based on the Intersection over Union (IoU) information [14] (IoU tracker) while the possessing relationship is predicted based on the position of the target object and the position of human keypoints in the neighborhood of the target object. The whole system runs at over 20 FPS and thus meets the real-time requirement.

The main contributions of this paper are two-fold:

- We propose an algorithm framework for relationship detection that is able to run in real-time.
- We test the proposed algorithm through several video sequences that are representative of the target application scenario (sports analytics) and validate its effectiveness.

2 Algorithm Structure

The whole algorithm consists of four parts: a visual object tracker, a human pose estimator, an IoU tracker, and a relationship reasoning module. Fig. 2 gives an overview.

2.1 Visual Object Tracker

In terms of interfaces, a visual object tracker provides two public methods, namely *initialize* and *update*. *initialize* accepts an initial frame as well as the localization of the target object on which tracking is performed, usually represented by its bounding box. At this point, *initialize* processes its inputs and initializes some internal state of the tracker. It can also be regarded as registration of template information. Later, *update* is called on every entry of a new frame and is fed with the image of this frame. The tracker processes this frame with the help of the registered template information as well as the tracking history information such as the position and size of the target object on the previous frame. Finally, the position and size of the target object are given by the tracker in the form of a bounding box. Note that the category of the target object is not pre-defined and can even be from a category not included in the training data.

In this paper, we adopt SiamFC++ [1] as our tracker. Its model structure is illustrated in Fig. 1. Taking as input a template patch and a search region patch, the siamese backbone network maps both images into feature space. The obtained feature maps, containing the template and search region information respectively, undergo a depth-wise cross-correlation operation [15] and a pair of feature maps after this template matching process. Then these feature maps are

This work is supported by the Key R&D Program of Zhejiang Province, China under Grant 2017C01007.

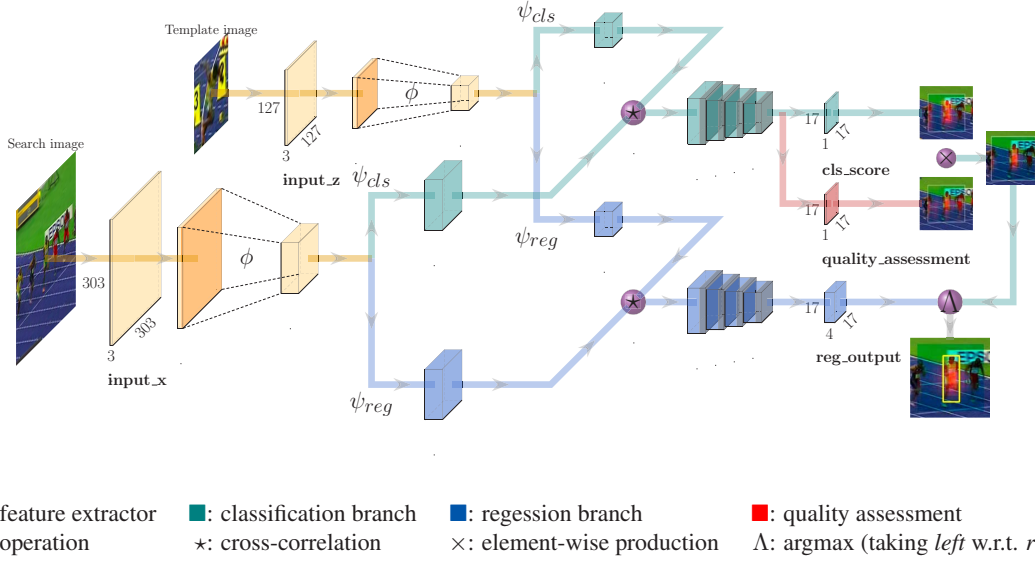


Fig. 1: Pipeline illustration of SiamFC++ [1] (AlexNet version). On the left side, the backbone network (in yellow) extracts features from both the template patch and the search region patch. In the middle of the figure, a matching process is performed over the extracted features. On the right side, from top to down, classification (in green), state (size) estimation quality assessment (in rose), and state (size) estimation (in blue) are performed and their results are exploited for the final prediction of target position and size.

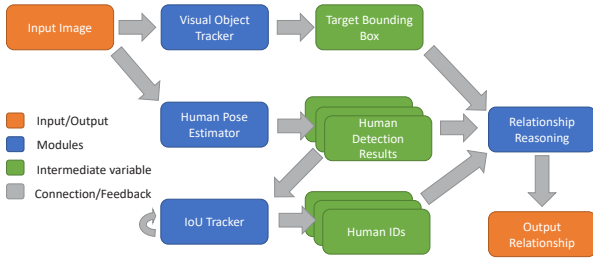


Fig. 2: Overview of the proposed algorithm for possessing relationship detection.

fed into a multi-branch head network for multi-task predictions:

- The classification aims at localizing the target object via an objectness score prediction.
- The size estimation quality assessment gives an estimation over the quality of the size regression and helps to select a feature map pixel with high regression quality.
- The target size regression regresses the distance between the center of its pixel and the border of the bounding box of the target.

We adopt a common post-processing method of scores and regression results which is widely used by the siamese tracking method (e.g. [15, 16]). Specifically, classification scores are corrected via multiplication with quality assessment scores. This score is then penalized based on continuous motion hypothesis and change of the regressed size. Finally, the pixel with the highest penalized score is chosen and its size regression result is smoothed with the prediction of the previous frame before being output.

Note that in the real implementation, the feature maps from the template patch that is used for template matching

are calculated only once and are reused through the video sequence.

2.2 Human Pose Estimator

Human Pose Estimation or Human Keypoint Detection task aims at detecting all visible instances of humans together with their joints (keypoints). In this paper, we adopt Objects as Points [10] as the human pose estimator of our algorithm. At first stage, with the Deep Layer Aggregation (DLA-34) [17] and the Deformable Convolutional Network (DCNv2) [18] whose architecture is illustrated in Fig. 3, visual features of input images are extracted and fused into one feature map with a downsampling factor of 4. Then this feature map is fed into a head network and a 39-dimensional vector is predicted at each pixel on the feature map. For a pixel $(\hat{x}_i, \hat{y}_i)$, this vector consist of:

- 1-D center point confidence (of being the center of a human), $\hat{Y}_{x_i y_i}$
- 2-D offset (horizontal/vertical) shared by both bounding box estimation and human pose estimation, $(\delta \hat{x}_i, \delta \hat{y}_i)$
- 2-D size (width/height), $(\hat{w}_i, \hat{h}_i)$
- 17x2-D offsets for keypoints in 17 joint categories, 2-D offsets (horizontal and vertical) per joint category, $(\delta \hat{x}_i^j, \delta \hat{y}_i^j), j = 1, 2, \dots, 17$

Based on the center point confidence $\hat{Y}_{x_i y_i}$, a set of human instances is selected and the pose and size information is decoded with the rest dimensions of the vector. The decoding process is described in Equation 1 and Equation 2:

$$(\hat{x}_i, \hat{y}_i) = (x_i, y_i) + (\delta \hat{x}_i, \delta \hat{y}_i) + (\delta \hat{x}_i^j, \delta \hat{y}_i^j), j = 1, 2, \dots, 17 \quad (1)$$

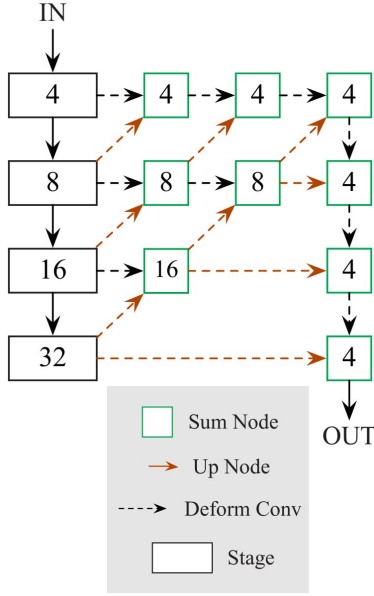


Fig. 3: Backbone architecture of Objects as Points [10].

$$(\hat{x}_i^{(lt)}, \hat{y}_i^{(lt)}, \hat{x}_i^{(rb)}, \hat{y}_i^{(rb)}) = (\hat{x}_i + \delta\hat{x}_i - \hat{w}_i/2, \hat{y}_i + \delta\hat{y}_i - \hat{h}_i/2, \hat{x}_i + \delta\hat{x}_i + \hat{w}_i/2, \hat{y}_i + \delta\hat{y}_i + \hat{h}_i/2) \quad (2)$$

where $(\hat{x}_i, \hat{y}_i)$ is the decoded keypoint coordinates and $(\hat{x}_i^{(lt)}, \hat{y}_i^{(lt)}, \hat{x}_i^{(rb)}, \hat{y}_i^{(rb)})$ is the decoded bounding box of human.

This estimator is able to be extended to other vision tasks such as 2D/3D object detection while we only concern its human pose estimation part in this paper.

2.3 IoU Tracker

While the human pose is estimated for each frame, it is still necessary to keep the person id. In this paper, we adopt IoU tracker [14] which is a simple yet effective and only requires the instance position information. Based on the continuity of human instance motion, this tracker uses IoU information to associate the incoming detection results (a set of bounding boxes) with the alive tracklets (a sequence of bounding boxes representing one person).

We adopt a slightly slimmed version of IoU tracker and the algorithm is presented in Algorithm 1.

2.4 Relationship Reasoning

Based on previously perceived information of the target object and the poses of present humans, a reasoning step is needed before achieving our goal of identifying the relationships.

In order to keep our whole pipeline able to run in real-time, we conceive a heuristic reasoning method that exploits the spatial information of instances. We assume that at any time there is at most one relation exists, which generally holds in sports games such as the frisbee and the football. The main idea is to choose the instance whose hand or foot has the highest centerness [20] with respect to an expanded version of the target bounding box. In this paper, we consider two types of relationships: "possessing by hand" (e.g.

Algorithm 1: IoU Tracker (slimmed)

```

Initialize (
    iou_threshold: threshold over IoU
)
Ta = ∅: // list of active tracklet
Register iou_threshold as hyper-parameter
Update (
    D = {Di}i=1L
)
target_ids ← {-1}i=1L // -1 is default value
// Assign existing tracklet to detection on current frame
for tracklet ta in Ta do
    bboxa ← first element of ta
    for all Di do
        IoUi ← IoU(Da, Di)
        i* ← arg maxi Di
    if IoUi* > iou_threshold then
        // Associated to an alive tracklet D = D \ Di
        Set target_ids with i*
    else
        // Death of tracklet
        Ta = Ta \ ta
// Birth of new tracklet
for all remaining Di ∈ D do
    Ta ← Ta ∪ Di
    Set target_ids with i
return target_ids

```

the frisbee) and "possessing by foot" (e.g. football).

The reasoning process is described in Algorithm 2. ω_H and ω_F are set to 5 and 1, respectively, based on the visualization result.

3 Implementation Details

In this section, we describe the technical details of our algorithm.

3.1 Visual Object Tracker

We employ the released model of SiamFC++[1] to track the target object. To ensure a , we choose the architecture with GoogLeNet [5] as its backbone. The tracker is trained on ILSVRC-VID/DET [21], COCO [22], Youtube-BB [23], LaSOT [24] and GOT-10k [25] and it takes around 6 hours for training with a batchsize of 256 on a PC with 8 2080Ti GPU.

3.2 Human Pose Estimator

We adopt the released model of CenterNet [10] with DLA-34 [17] backbone as our human pose estimator. The estimator is trained on COCO [22] dataset with its annotation of bounding boxes and keypoints. According to [10], it takes 2.5 days to train on 8 GPUs.

4 Results

4.1 Experiment Settings

We give qualitative results over public video sequences. Our experiment involves to relationship types, namely *possessing by hand* (e.g. frisbee) and *possessing by foot* (e.g. football). Our algorithm runs at a speed of over 20 FPS on a PC with NVIDIA GeForce 1080Ti GPU, which makes it meet the real-time requirement of analyzing simultaneously.



Fig. 4: Visualization of our algorithm. Row 1 is the sequence *frisbee* from VOT2018 [19], while row 2-6 are public video sequences from Youtube². The first column presents corresponding targets (frisbee/football), the second column gives an overview of the scenario, and the last 4 column illustrates some details of the interaction between the target and the holder. Challenges include the fast motion and occlusion of the target object as well as the presence of different distractors. Better viewed while zoomed in.

4.2 Qualitative Results

We test our algorithm on videos from VOT, a benchmark for visual object tracking, and public videos from Youtube. From the visualization results in Fig. 4, the algorithm is capable to handle most of the cases by successfully return the low-level position information (target bounding box and human poses) and the high-level relationship (possessing by hand/foot).

One of the challenges comes from the fast motion of the

target object (both for frisbee and football), which indicates that the target object may be partially out of the search region and be blurred. Owing to the abundance of training data described in Section 3.1, the VOT tracker can learn a visual representation that is capable to localize the target object despite the fast motion.

Another challenge lies in the occlusion (by hand/foot) and distractors (the body part of the players) as the interaction between the target object and the players can be very intensive in our application scenario. Again, thanks to the diversity of the training data, the visual representation learned by

²frisbee: <https://m.youtube.com/watch?v=f7h5nNyH8M>
football: <https://m.youtube.com/watch?v=bKOTKhtbM54>

Algorithm 2: Fast Relationship Reasoning Algorithm

Input:

$\text{bbox}_{\text{target}}$: target bounding box
 $\mathbf{K} = \{K_i^{\text{LH}}, K_i^{\text{RH}}, K_i^{\text{LF}}, K_i^{\text{RF}}\}_{i=1}^L$: human keypoints
(left hand, right hand, left foot, right foot, respectively)
 α : expansion ratio

Output:

id: id of person with which the target has relationship R:
relationship type
Hyper-Parameter(
 ω_H : weight of hand parts for calculation of ctr_score
 ω_F : weight of foot parts for calculation of ctr_score)

```
Expand  $\text{bbox}_{\text{target}}$ .  
// ctr: center-ness  
 $\text{part\_ctr\_score}_i^P \leftarrow \text{ctr}(K_i^P, \text{bbox}_{\text{target}})$   
 $\text{part\_ctr\_score}_i^H \leftarrow$   
   $\max(\text{part\_ctr\_score}_i^{\text{LH}}, \text{part\_ctr\_score}_i^{\text{RH}})$   
 $\text{part\_ctr\_score}_i^F \leftarrow$   
   $\max(\text{part\_ctr\_score}_i^{\text{LF}}, \text{part\_ctr\_score}_i^{\text{RF}})$   
 $\text{ctr\_score}_i \leftarrow \omega_H \cdot \text{part\_ctr\_score}_i^H + \omega_F \cdot \text{part\_ctr\_score}_i^F$   
 $i^* \leftarrow \arg \max_i [\text{ctr\_score}_i]$   
if  $\text{ctr\_score}_{i^*} > 0$  then  
  id =  $i^*$   
  if  $\max(K_{i^*}^{\text{LH}}, K_{i^*}^{\text{RH}}) \geq \max(K_{i^*}^{\text{LF}}, K_{i^*}^{\text{RF}})$  then  
    relation type = 'possessing by hand'  
  else  
    relation type = 'possessing by foot'  
else  
  id = -1 // no matched person  
return id, R
```

our VOT tracker is robust enough to track the target with the presence of distractor and occlusion.

4.3 Analysis and Future Improvements

While the algorithm framework has been validated via qualitative results, the following aspects require more attention during future researches:

- **Continuity for human pose prediction** The continuity for human pose prediction need to be taken into consideration so that both prediction quality and robustness can be improved. One solution can be to exploit video data with pose annotation during training [26].
- **Robustness of the person tracker** In this paper, a simple IoU tracker is enough for validation of the framework. However, scenarios of real applications can be more complicated and a more sophisticated person tracker will be needed in this case. For example, literature and methods from the field of Articulated Multi-person Tracking [27, 28] shall provide insights for this issue.
- **Reasoning rules for Relationship** In this paper, we only consider two relationships and their reasoning rules are straight-forward. Yet to build a system capable of real application, it may involve both more relationships and more robust reasoning rules/methods.

5 Conclusion

In this paper, we propose an algorithm framework for relationship detection in sports analytics. It exploits a visual object tracker to track the target object and utilizes a human pose estimator to predict the human poses. With an IoU tracker, the identity information of persons is preserved. Finally, a relationship reasoning is performed based on information from previous steps. The visualization results demonstrate the effectiveness of our algorithm.

References

- [1] Y. Xu, Z. Wang, Z. Li, Y. Ye, and G. Yu, "Siamfc++: Towards robust and accurate visual tracking with target estimation guidelines," *arXiv preprint arXiv:1911.06188*, 2019.
- [2] J. Van Haaren, A. Zimmermann, J. Renkens, G. Van den Broeck, T. Op, D. Beéck, W. Meert, and J. Davis, "Machine learning and data mining for sports analytics," 2013.
- [3] R. P. Bunker and F. Thabtah, "A machine learning framework for sport result prediction," *Applied computing and informatics*, vol. 15, no. 1, pp. 27–33, 2019.
- [4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- [5] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2818–2826, 2016.
- [6] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "Shufflenet v2: Practical guidelines for efficient cnn architecture design," in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 116–131, 2018.
- [7] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," in *Advances in neural information processing systems*, pp. 91–99, 2015.
- [8] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," *arXiv preprint arXiv:1706.05587*, 2017.
- [9] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "Arcface: Additive angular margin loss for deep face recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4690–4699, 2019.
- [10] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," *arXiv preprint arXiv:1904.07850*, 2019.
- [11] G. Gkioxari, R. Girshick, P. Dollár, and K. He, "Detecting and recognizing human-object interactions," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 8359–8367, 2018.
- [12] K. Kato, Y. Li, and A. Gupta, "Compositional learning for human object interaction," in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 234–251, 2018.
- [13] H.-S. Fang, J. Cao, Y.-W. Tai, and C. Lu, "Pairwise body-part attention for recognizing human-object interactions," in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 51–67, 2018.
- [14] E. Bochinski, V. Eiselein, and T. Sikora, "High-speed tracking-by-detection without using image information," in *2017 14th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, pp. 1–6, IEEE, 2017.
- [15] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, and J. Yan, "Siamrpn++: Evolution of siamese visual tracking with very deep networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4282–4291, 2019.

- [16] B. Li, J. Yan, W. Wu, Z. Zhu, and X. Hu, "High performance visual tracking with siamese region proposal network," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 8971–8980, 2018.
- [17] F. Yu, D. Wang, E. Shelhamer, and T. Darrell, "Deep layer aggregation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2403–2412, 2018.
- [18] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable convnets v2: More deformable, better results," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 9308–9316, 2019.
- [19] M. Kristan, A. Leonardis, J. Matas, M. Felsberg, R. Pflugfelder, L. Cehovin Zajc, T. Vojir, G. Bhat, A. Lukežić, A. Eldesokey, *et al.*, "The sixth visual object tracking vot2018 challenge results," in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 0–0, 2018.
- [20] Z. Tian, C. Shen, H. Chen, and T. He, "Fcos: Fully convolutional one-stage object detection," in *Proceedings of the IEEE International Conference on Computer Vision*, pp. 9627–9636, 2019.
- [21] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, *et al.*, "Imagenet large scale visual recognition challenge," *International journal of computer vision*, vol. 115, no. 3, pp. 211–252, 2015.
- [22] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *European conference on computer vision*, pp. 740–755, Springer, 2014.
- [23] E. Real, J. Shlens, S. Mazzocchi, X. Pan, and V. Vanhoucke, "Youtube-boundingboxes: A large high-precision human-annotated data set for object detection in video," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 5296–5305, 2017.
- [24] H. Fan, L. Lin, F. Yang, P. Chu, G. Deng, S. Yu, H. Bai, Y. Xu, C. Liao, and H. Ling, "Lasot: A high-quality benchmark for large-scale single object tracking," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 5374–5383, 2019.
- [25] L. Huang, X. Zhao, and K. Huang, "Got-10k: A large high-diversity benchmark for generic object tracking in the wild," *arXiv preprint arXiv:1810.11981*, 2018.
- [26] U. Iqbal, A. Milan, and J. Gall, "Posetrack: Joint multi-person pose estimation and tracking," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 2011–2020, 2017.
- [27] E. Insafutdinov, M. Andriluka, L. Pishchulin, S. Tang, E. Levinkov, B. Andres, and B. Schiele, "Artrack: Articulated multi-person tracking in the wild," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 6457–6465, 2017.
- [28] S. Jin, W. Liu, W. Ouyang, and C. Qian, "Multi-person articulated tracking with spatial and temporal embeddings," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 5664–5673, 2019.